

KVERNERGET KRISTIANSUND, Norway

WMO: 012230

Lat: 63.1094N

Lon: 7.8158E

Elev: 62

StdP: 100.58

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.7	-13.4	1.2	-4.5	-11.8	1.4	-3.4	14.6	4.6	13.1	4.9	3.3	90	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	23.8	16.2	21.4	15.1	19.6	14.3	17.2	22.1	15.9	20.0	15.0	18.3	3.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	10.7	18.6	14.1	10.1	17.5	13.3	9.6	16.4	48.3	22.3	44.9	20.2	42.2	18.2	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.9	10.9	9.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-8.2	27.5	2.7	2.3	-10.1	29.1	-11.7	30.5	-13.2	31.7	-15.1	33.4		
			-9.1	19.0	2.4	1.4	-10.8	20.0	-12.2	20.8	-13.6	21.6	-15.3	22.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	1.5	2.0	3.1	6.0	9.2	11.6	14.4	12.1	7.9	4.2	2.4	
	DBStd	5.65	3.24	3.34	2.92	3.33	3.63	2.83	3.13	2.52	2.66	3.03	3.53	3.61
	HDD10.0	1403	265	224	216	129	58	12	1	1	9	78	174	236
	HDD18.3	3996	523	458	474	371	283	203	131	124	188	323	423	494
	CDD10.0	463	0	0	0	8	34	60	137	138	72	13	1	0
	CDD18.3	14	0	0	0	0	1	1	9	4	0	0	0	0
	CDH23.3	79	0	0	0	0	3	6	58	12	0	0	0	0
	CDH26.7	8	0	0	0	0	0	0	8	0	0	0	0	0
Wind	WSAvg	3.9	4.1	4.0	4.6	3.7	3.8	4.1	3.6	3.2	3.8	3.8	3.9	4.3
Precipitation	PrecAvg	1635	172	150	144	88	80	105	107	136	153	152	152	196
	PrecMax	1965	421	297	316	261	148	184	178	274	392	280	382	380
	PrecMin	1105	5	9	49	24	33	35	43	43	56	33	10	51
	PrecStd	207	95	72	72	54	27	40	35	54	81	56	100	80
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.4	11.3	19.6	23.4	24.1	27.8	25.0	20.8	17.7	13.1	11.4
		MCWB	5.7	6.3	5.7	11.1	13.8	15.7	18.9	16.6	14.7	11.4	9.5	7.9
	2%	DB	8.1	8.7	9.5	15.9	19.9	20.9	25.0	22.5	18.9	15.8	11.5	9.5
		MCWB	5.1	5.1	5.3	9.1	12.7	14.2	17.7	15.4	13.2	10.3	8.1	6.4
	5%	DB	7.1	7.5	8.3	12.9	17.2	18.1	22.0	20.2	17.2	13.3	10.1	8.3
		MCWB	4.7	4.4	5.1	7.9	11.6	13.0	16.3	14.8	12.9	9.1	7.1	5.6
	10%	DB	5.9	6.5	7.2	10.8	15.0	16.1	19.4	18.6	16.0	12.0	8.9	7.1
		MCWB	3.8	3.8	4.5	6.5	10.2	12.2	15.2	14.5	12.3	8.7	6.3	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	7.6	7.8	11.6	14.9	16.1	19.7	17.5	15.7	12.4	10.3	8.9
		MCDB	8.3	9.3	8.8	19.0	21.4	22.8	26.5	22.4	18.9	15.0	12.2	10.3
	2%	WB	6.2	6.1	6.9	9.6	13.2	14.6	18.1	16.3	14.5	11.1	8.9	7.5
		MCDB	7.5	7.5	8.2	14.9	18.7	19.5	24.1	20.4	17.0	14.1	10.8	8.7
	5%	WB	5.0	5.2	5.7	8.3	12.0	13.5	16.7	15.5	13.6	10.0	7.7	6.0
		MCDB	6.2	6.7	7.6	11.8	16.8	17.4	21.1	18.9	16.0	13.1	9.4	7.8
	10%	WB	3.9	4.1	4.7	7.2	10.7	12.5	15.5	14.7	12.8	9.0	6.6	4.7
		MCDB	5.7	6.0	6.7	9.6	14.5	15.7	18.9	17.7	15.3	11.3	8.4	6.4
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.3	5.3	6.1	6.6	5.6	5.8	5.5	5.0	4.1	4.0	
		MCDBR	5.0	5.4	6.4	9.4	10.4	9.2	10.5	8.9	7.1	6.5	5.3	5.7
		MCWBR	3.7	3.6	4.1	4.7	5.1	4.5	4.7	4.3	3.8	3.9	3.6	4.4
	5% WB	MCDBR	4.6	4.7	5.5	8.3	9.9	8.3	9.8	7.4	6.4	6.0	5.1	5.6
		MCWBR	3.8	3.8	4.1	4.5	5.0	4.4	4.7	3.9	3.8	4.0	4.0	4.8
Clear-Sky Solar Irradiance	taub	0.242	0.300	0.327	0.349	0.348	0.350	0.358	0.349	0.332	0.308	0.375	0.357	
	taud	2.073	2.266	2.333	2.388	2.457	2.487	2.487	2.512	2.541	2.451	2.708	2.567	
	Ebn at Noon	383	639	772	834	869	873	854	831	776	640	371	208	
	Edh at Noon	31	59	81	95	96	95	93	83	67	49	28	18	
All-Sky Solar Radiation	RadAvg	0.15	0.63	1.81	3.51	4.77	5.03	4.63	3.56	2.11	0.93	0.23	0.08	
	RadStd	0.02	0.12	0.27	0.32	0.25	0.48	0.41	0.33	0.25	0.14	0.05	0.01	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.43	+0.67	N/A	N/A	N/A	-139	-183	+58

Nomenclature: See separate page